WEBLINKS TO TRADE

Estimating Interregional Trade using Website Information

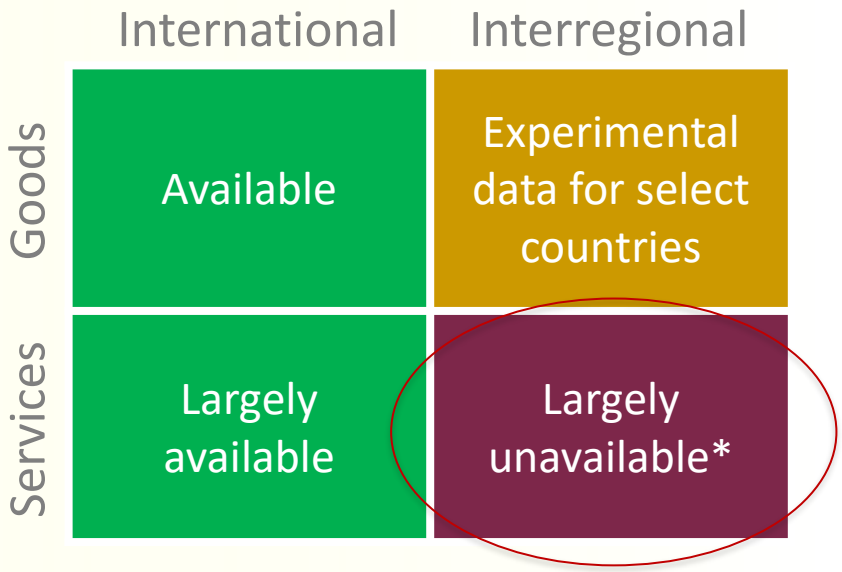
Network Science & Economics
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OECD
BETTER POLICIES FOR BETTER LIVES

Measuring interregional trade

Available trade data



Our Contribution

- Create a scalable, bottom-up method based on publicly available data to proxy trade between regions.
- Based on connections between firm websites with firm-level characteristics.
  - Use a pilot project in the UK pilot, benchmark with micro-data.
  - Allign with recent academic research (e.g. Tranos et al., 2023).
- Our contribution is to create micro-founded estimates with a focus on trade in services.

Note: * No official region-to-region trade flows between Member countries or countries within the European Union.
Selected current and past efforts by individual Member countries, e.g., United Kingdom and Finland.

Existing approaches to measure interregional trade



1. Trade surveys from national statistics agencies. Collected by a handful of countries e.g. Wales, Scotland, Finland, Netherlands. Time consuming, low response rates, long time horizons.



2. Administrative data and other proxy data. Proxies for trade using administrative data: shipping and freight data, VAT records, business-to-business transactions. Spotty availability across countries, better coverage for goods.



Our measure uses open website data with timely availability across-countries.

How do we create our data?

Constructing our data

Matching Orbis and CommonCrawl

Orbis

Firm Data

- Balance sheet data, company URL, and address.
- Keep economically active firms in period 2017-2021.

Common Crawl

Weblink Data

- Crawls of the public web.
- Weblink pairs with an origin and destination website.

Match exact string for *origin* and *destination* based on URL

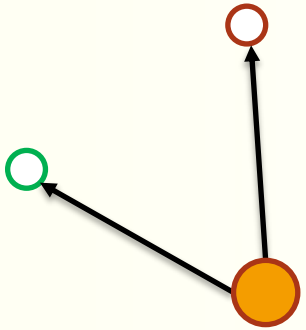
Matched firm pairs

www.astrazeneca.co.uk/about-us | **uk.co.astrazeneca**

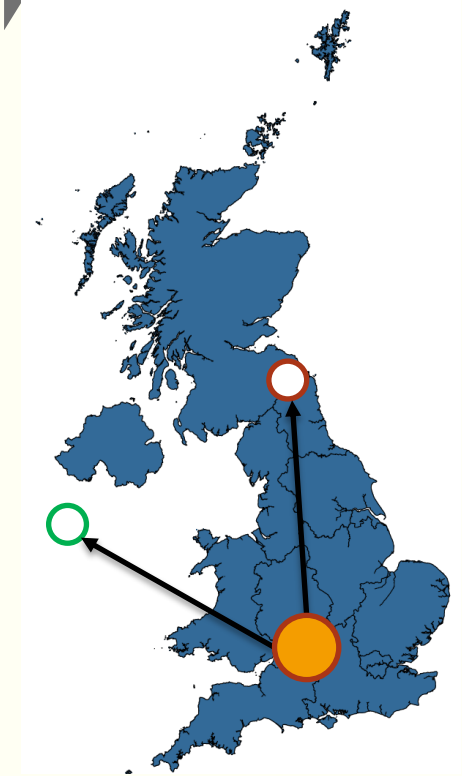
locations.

Implementation for United Kingdom

Hyperlink graphs identify website connections



Implementation for United Kingdom

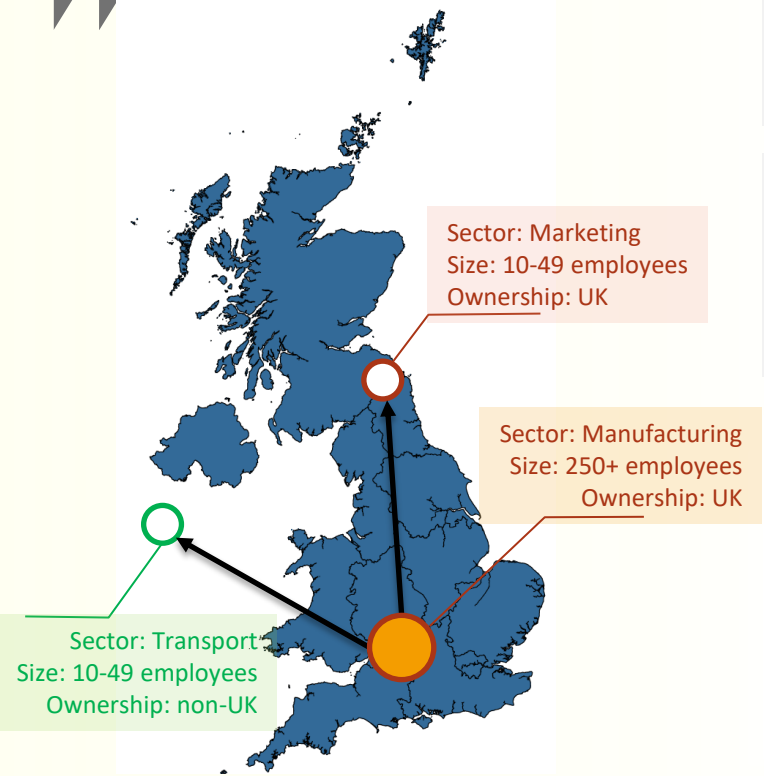


Hyperlink graphs identify website connections

Orbis provides firm-level information

- Geographical location (e.g., TL2, etc.)

Implementation for United Kingdom

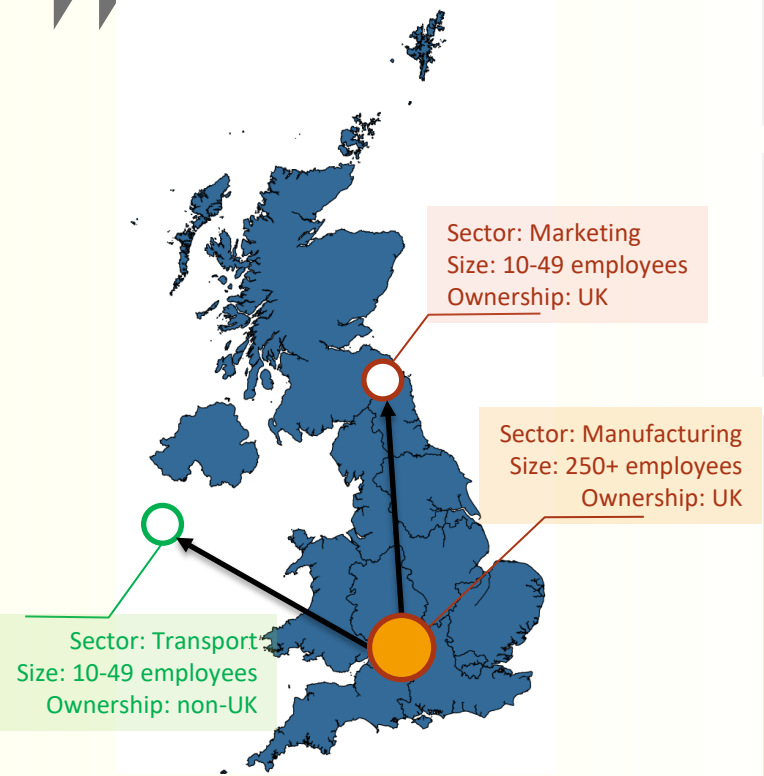


Hyperlink graphs identify website connections

Orbis provides firm-level information

- Geographical location (e.g., TL2, etc.)
- Firm-level characteristics (e.g., sector, size, etc.)

Implementation for United Kingdom



Hyperlink graphs identify website connections

Orbis provides firm-level information

- Geographical location (e.g., TL2, etc.)
- Firm-level characteristics (e.g., sector, size, etc.)

We match 835k (out of 1.4M) UK firms with a website at the TL3 level

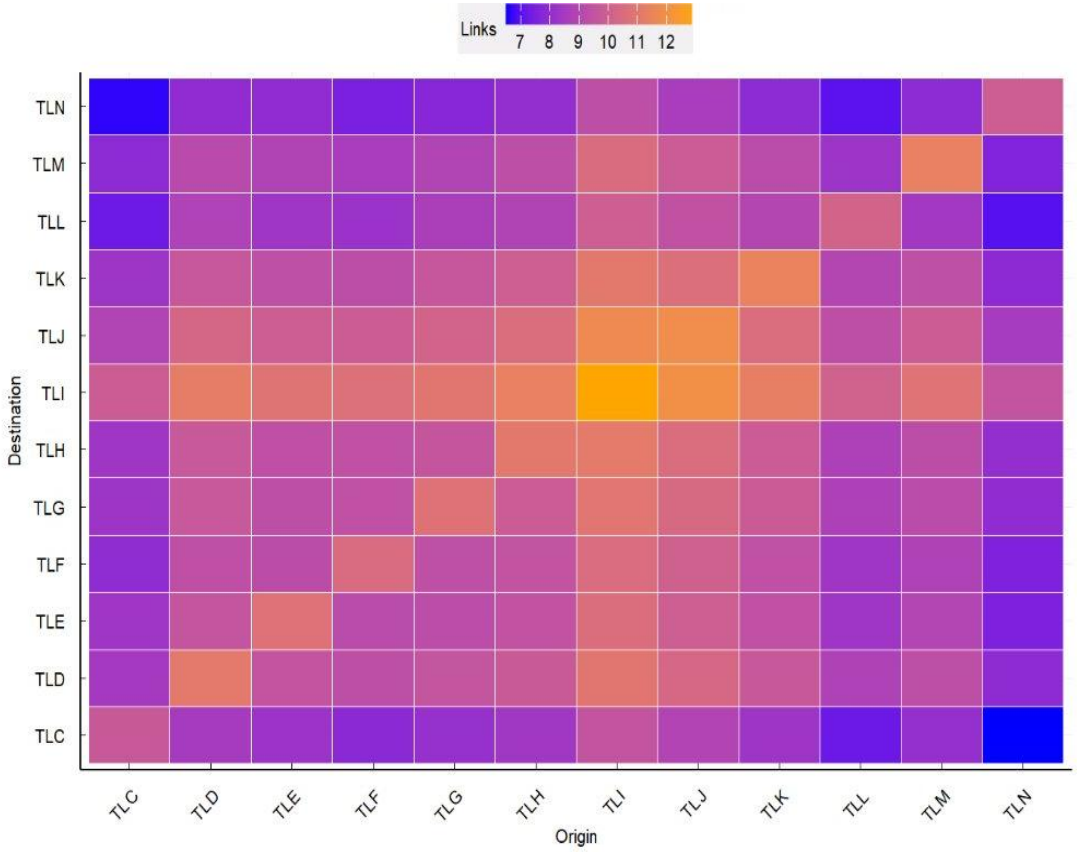
- 3.5 million interregional connections within the United Kingdom.
- Expanded coverage to include Irish firms.

Constructing our data

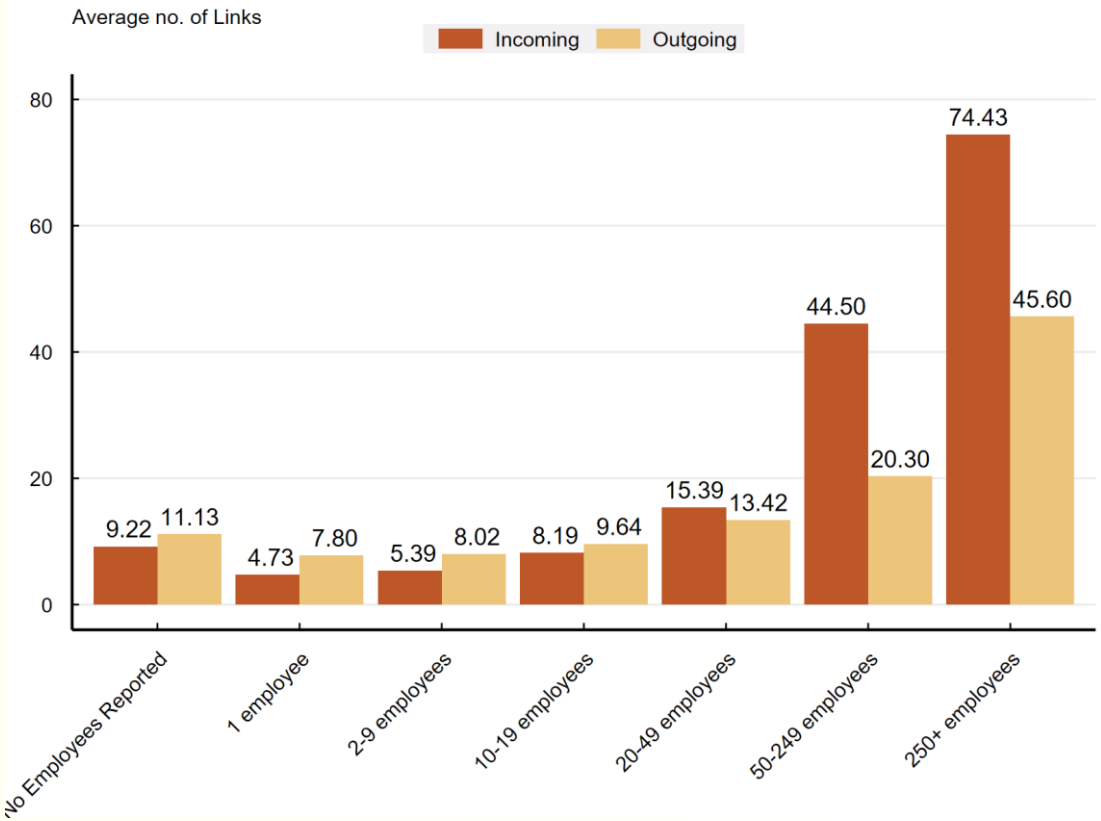
Matched firm-to-firm data

- We observe the characteristics of the origin and destination firms in the matched data.
  - We aggregate to the TL3, TL2 region and NACE sector.

- Heatmap of the origin and destination TL2 region.



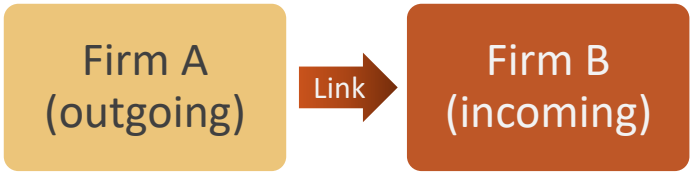
The CC data



We identify ‘incoming’ and ‘outgoing’ links.

The average firm has 7 incoming and 9 outgoing weblinks.

Incoming links scale much faster than outgoing.



How balanced is the sample?

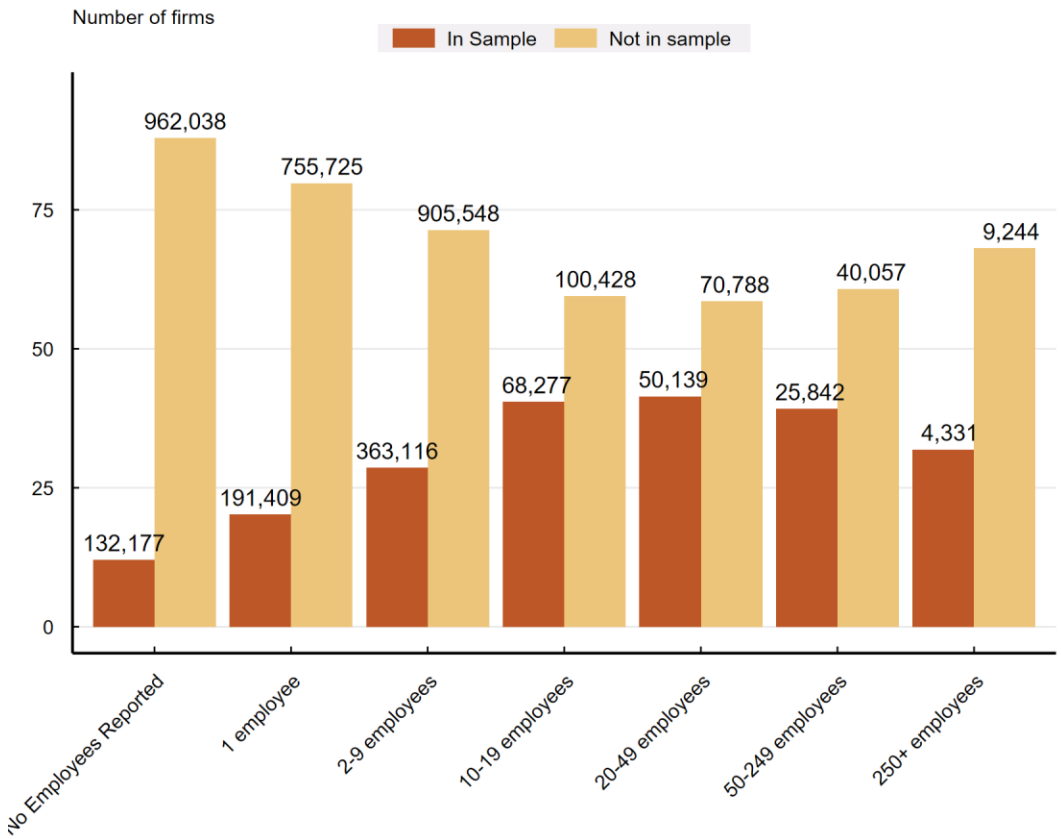
Balance table, median values (2019)

	Matched	Unmatched	Overall
No. Firms	835,291	2,843,828	3,679,119
Employees	3	2	2
Assets (£)	96,292	28,884	36,491
Revenue (£)	214,353	178,300	194,667
Export Revenue (£)	3,761,433	4,009,991	3,924,778
R&D Expense (£)	251,356	356,353	301,522
Age	10	4	6

- The median firm in our sample …
  - is larger in terms of assets and employees.
  - Is older
  - Earns slightly less in operating revenue
  - Reports lower R&D expenses and revenue from exports.

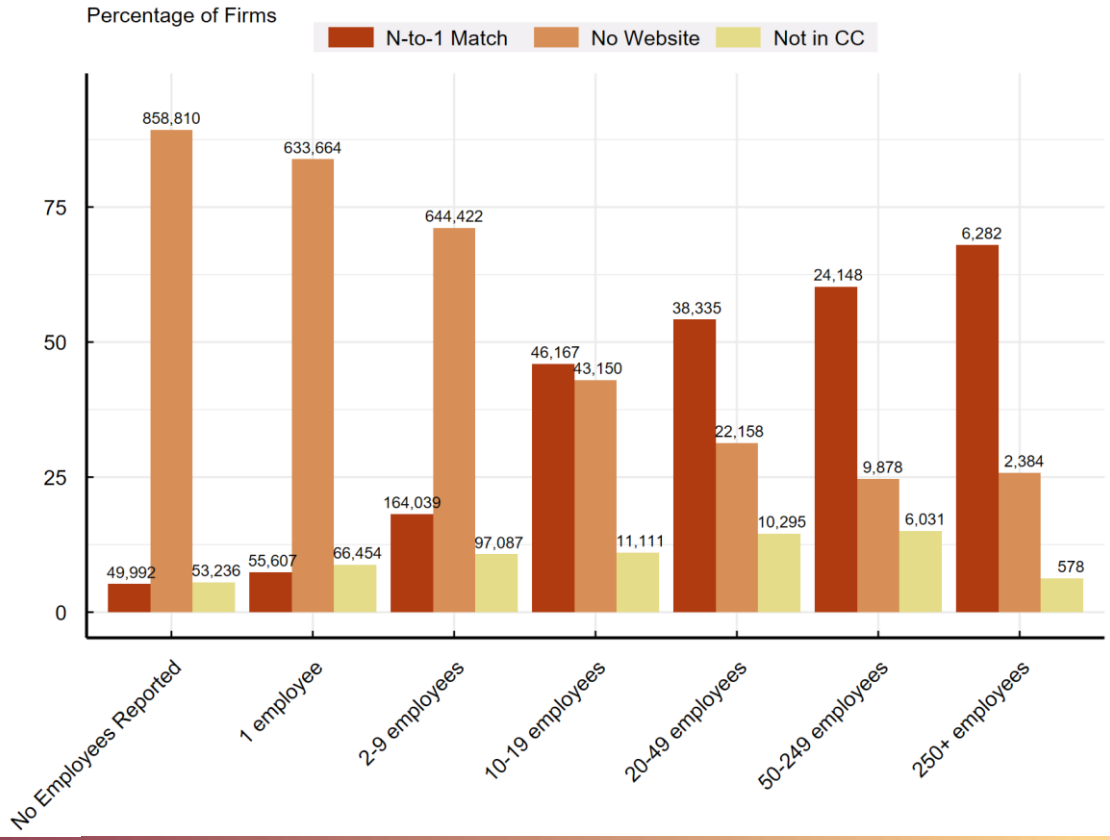
Sampling

- The match rate varies by firm size:
  - Very few single or zero employee firms are in our sample.
  - We match many more medium-sized firms.
  - But this declines for 250+ employee firms.



Why do we miss firms?

- >75% do not have a website in Orbis (<10 employees)
- We do not match some larger firms due to ‘n-to-1’ matches.
- Large firms often report multiple entities in Orbis. For now, we exclude these.



Benchmarking our approach

Benchmarking

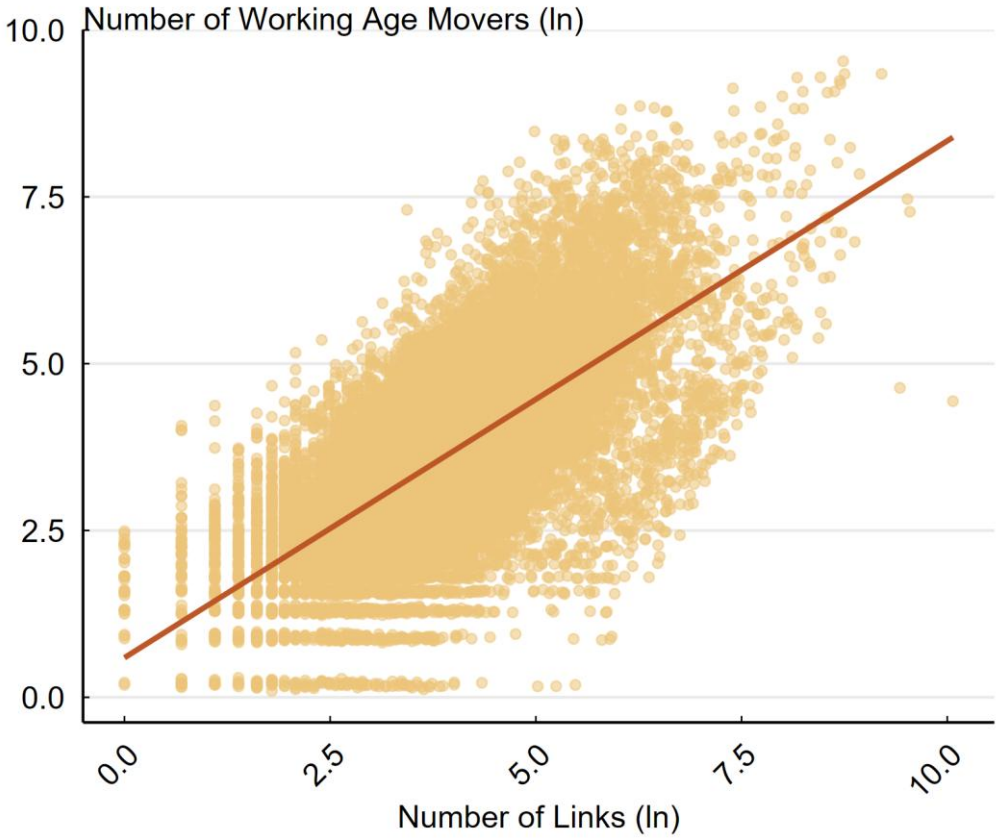
Compare against other proxy approaches for measuring trade. Test how well the weblink data correlates with **origin-destination regional migration**, **EUREGIO** trade data and **survey-based trade estimates** from the UK.

Replicate existing academic research using proxy approaches for inter-regional trade in **Santamaria et al. (2023)**. Compare effect sizes between approaches.

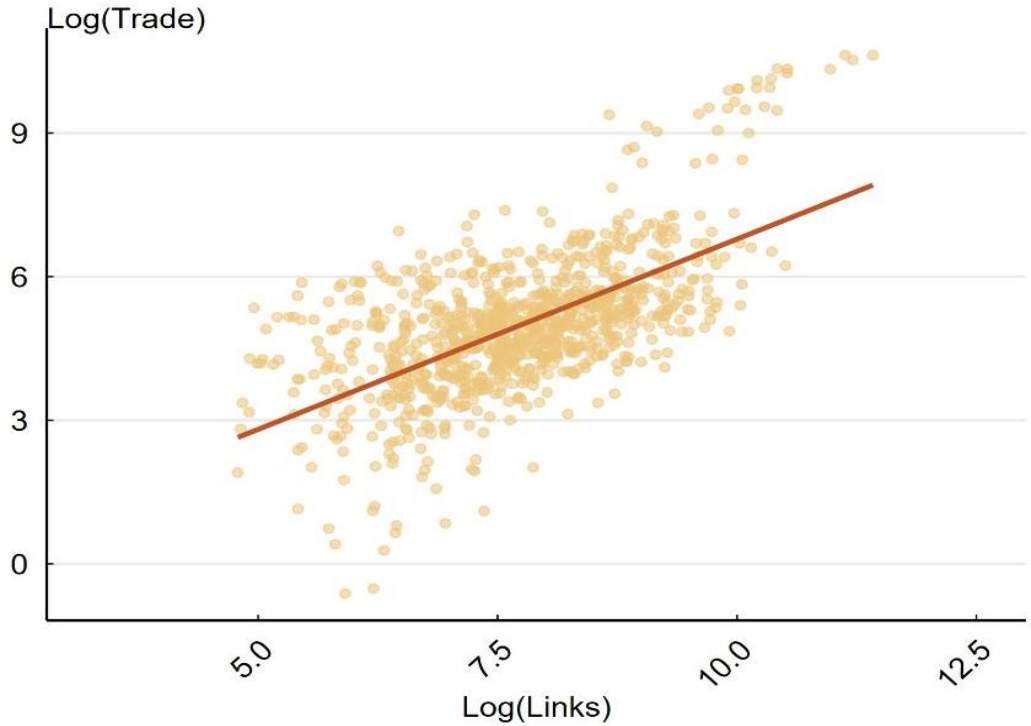
Origin-destination migration data

- Migration is often used as a proxy for trade.
  - We compare the weblinks to origin-destination regional migration in England.

- Weblinks explains 54% of the variation in regional migration.

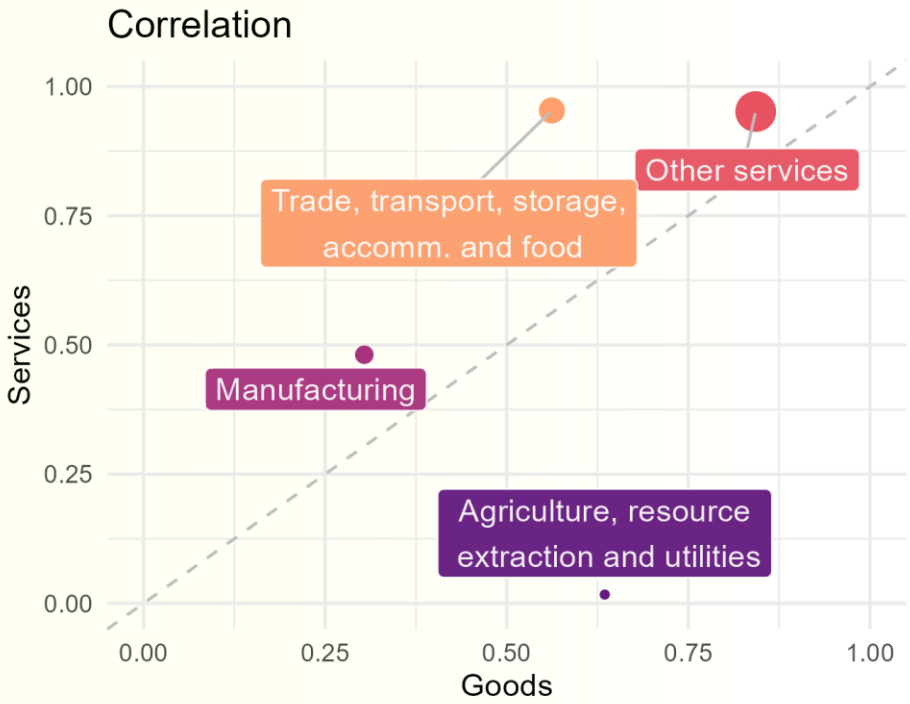


EUREGIO trade



- Weblinks explains 39% of the variation in EUREGIO data, 81% once controlling for the sending and receiving region.
- Comparisons should be treated with caution as the latest date in EUREGIO is 2010.
  - Points to the demand for more timely regional trade data!

Benchmarking weblinks and international trade



How much of the variation of international goods and services exports of UK regions can be captured using weblinks?

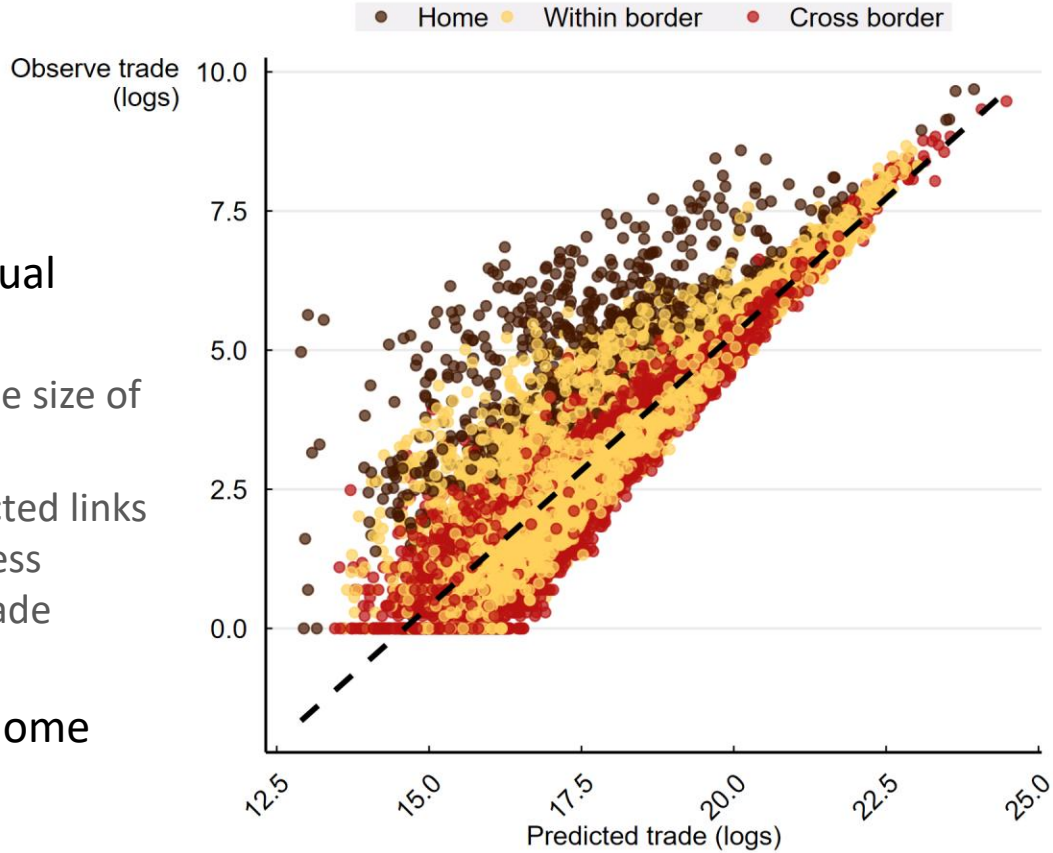
- More than 95% of the variation in service exports from service sectors
- Less than 30% of the variation in goods exports in the manufacturing sector

Weblinks might be a suitable proxy for services exports (and interregional services trade) but not for goods trade

Home bias in weblinks

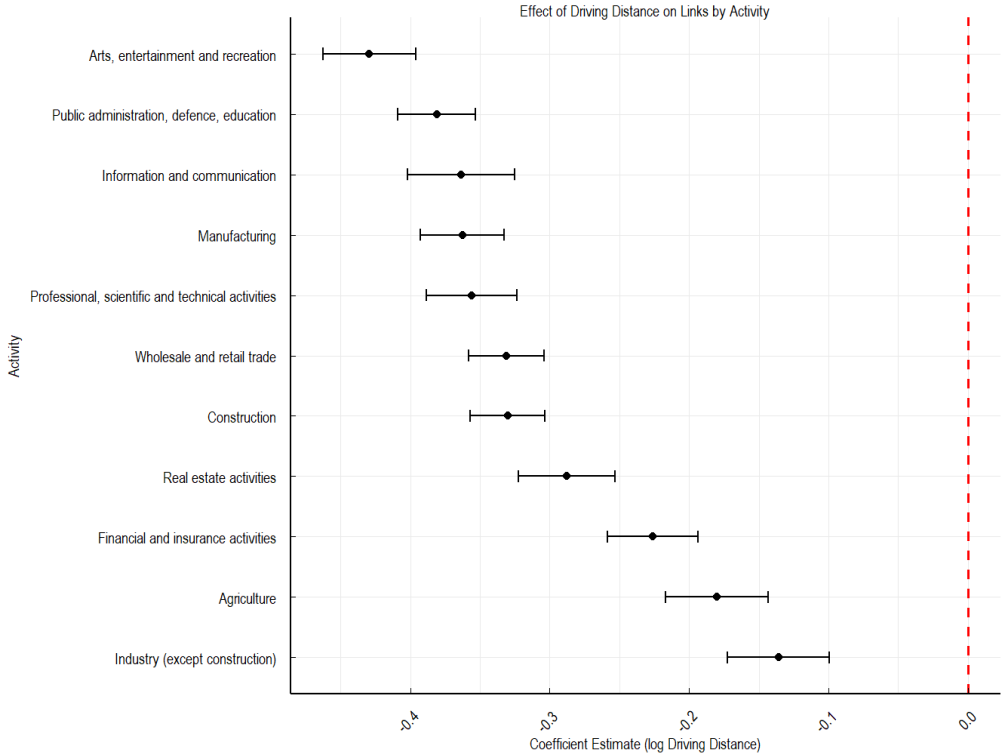
TL3 pairs

- Replicate existing work from Santamaria et al. (2023).
- We create a set of predicted and actual weblinks between TL3 regions.
  - Predicted trade is created by only the size of the origin and destination region.
  - Systematic variation from the predicted links suggest the influence of market access barriers or other non-size related trade preferences.
- We show descriptive evidence of a home bias in weblinks.



Gravity model findings: Distance

TL3 pairs, subsample by activity

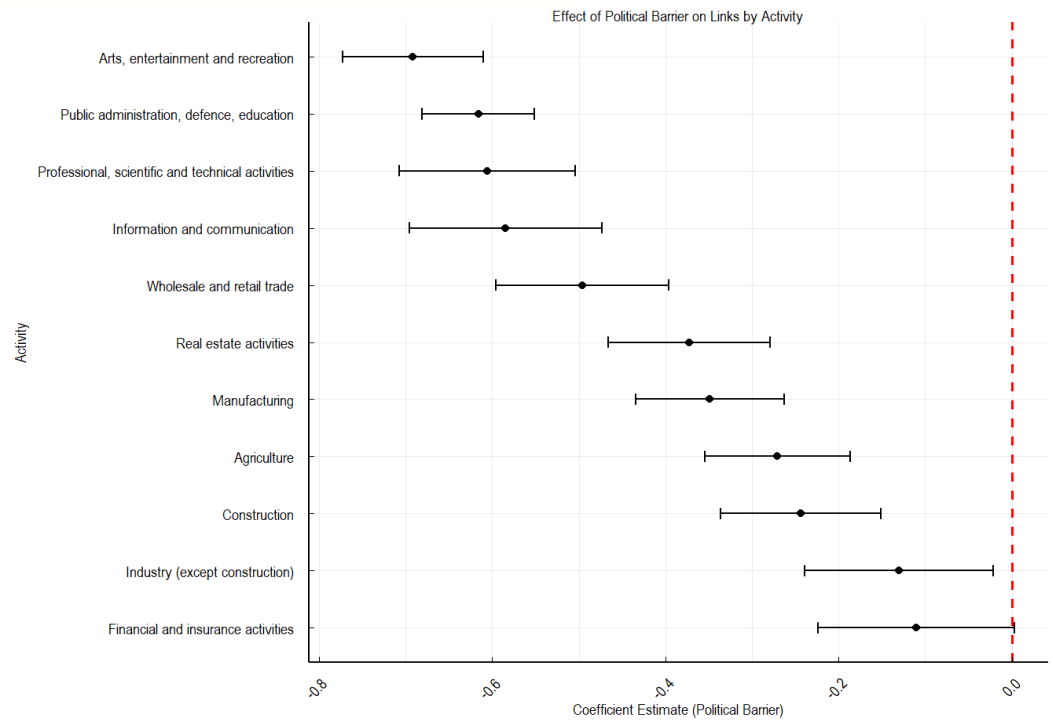


Most sensitive to distance:
Arts, entertainment and recreation;
Public administration, defence education

Least sensitive to distance:
Financial and insurance;
Agriculture;
Industry (except construction)

Gravity model findings: Political Borders

TL3 pairs, subsample by activity



Most sensitive to borders:
Arts, entertainment and recreation;
Public administration, defence education

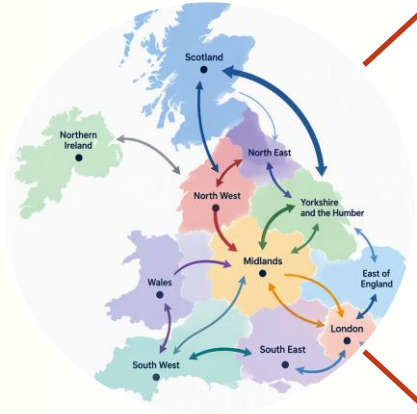
Least sensitive to borders:
Financial and insurance;
Industry (except construction)

What can the data tell us?

Estimating the trade part of the UK's MRIOs

ONS' experimental estimates

- Comprehensive data sources
- Less granular and missing industries



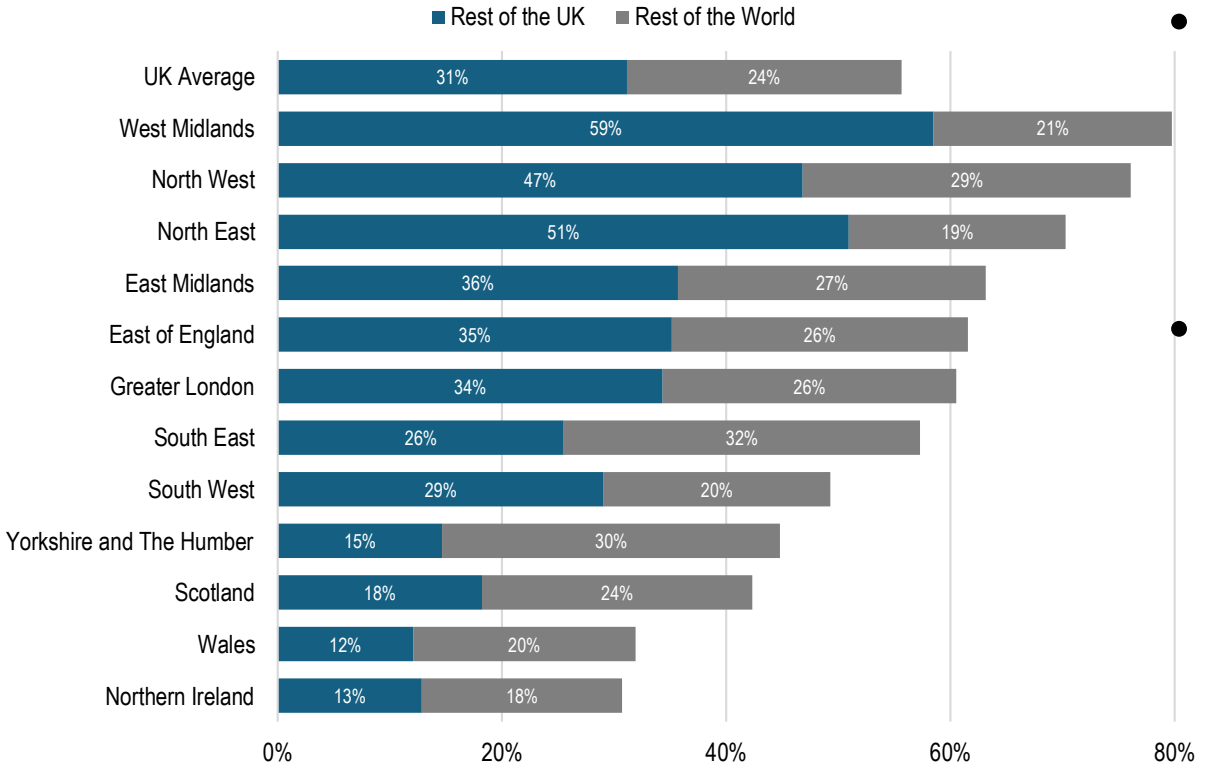
Weblinks

- Any target granularity available
- Missing MREs

BRES

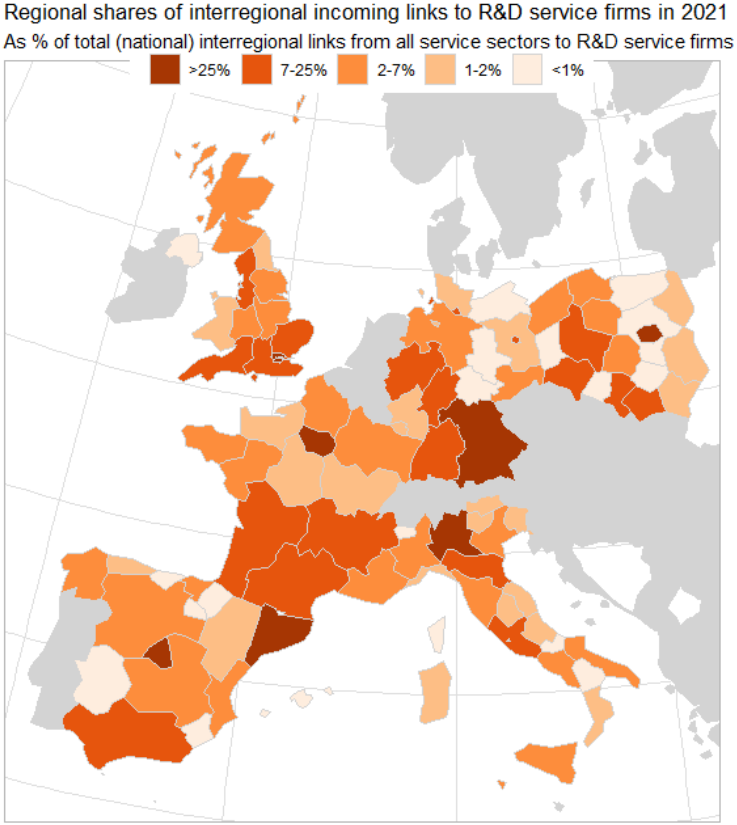
- Full universe of MREs

Origin of value added embodied in UK regions' final demand, 2019



- Over 55% of GVA originates outside of the region
- Interregional flows are more important in the majority of regions

Regional (TL2) distribution of incoming links to R&D intensive service firms



The R&D intensive service networks are **highly concentrated** in most countries (except 2 centres in Spain)

- For example, London accounts for:
  - 13% of UK's population in 2021
  - 23% of UK's GDP in 2020
  - 29% of UK's R&D intensive service firms in 2018
  - **40% interregional incoming links** (from UK services) to R&D intensive service firms in 2021

Utilisation and next steps

- Improve data processing
  - Matching for multi-unit firms
  - Improve handling of headquarters bias

- Link with micro-data
  - Benchmark and train weblink data on UK inter-regional financial transaction data

Any comments or suggestions are welcome!

Thank you!

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Santamaría, M., Ventura, J., & Yeşilbayraktar, U. (2023). Exploring European regional trade. *Journal of International Economics, 146*, 103747–103747.

Thissen, M., Lankhuizen, M., van Oort, F., Los, B., & Diodato, D. (2018). EUREGIO: The Construction of a Global IO Database with Regional Detail for Europe for 2000–2010; *Tinbergen Institute Discussion Paper*; No. TI 2018–084/VI.

Tranos, E., Carrascal-Incera, A., & Willis, G. L. (2022). Using the Web to Predict Regional Trade Flows: Data Extraction, Modeling, and Validation. *Annals of the American Association of Geographers, 113*(3), 717–739.

Appendix

Gravity estimates for weblinks

Fixed effects regression for TL3 pairs

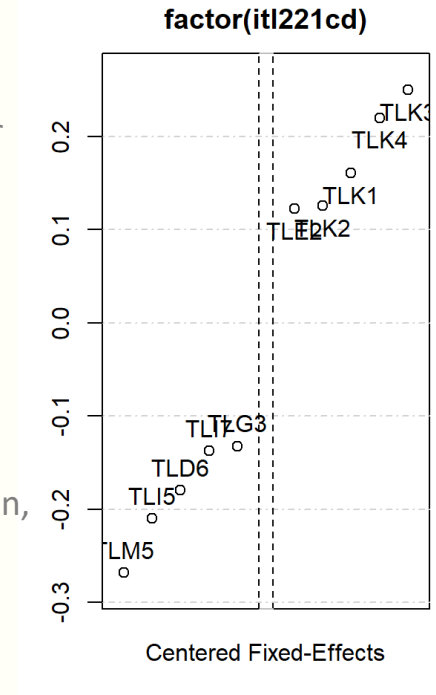
	Links (ln)			
Distance, km (ln)	-0.5198***	-0.3197***		
	(0.0247)	(0.0140)		
Driving Distance, km (ln)			-0.3304***	
			(0.0145)	
Travel Time, minutes (ln)				-0.3924***
				(0.0154)
Border	-0.7018***	-0.6065***	-0.6006***	-0.5799***
	(0.0233)	(0.0386)	(0.0400)	(0.0421)
Home	-0.1503	1.013***	0.8679***	0.6404***
	(0.1435)	(0.0600)	(0.0617)	(0.0611)
Year FE	Yes	Yes	Yes	Yes
Origin FE	No	Yes	Yes	Yes
Destination FE	No	Yes	Yes	Yes
Observations	128,661	128,661	128,661	128,661
R^2	0.29199	0.87351	0.87395	0.87400
Within R^2	0.21250	0.32445	0.32678	0.32707
Note:				*p<0.1; **p<0.05; ***p<0.01

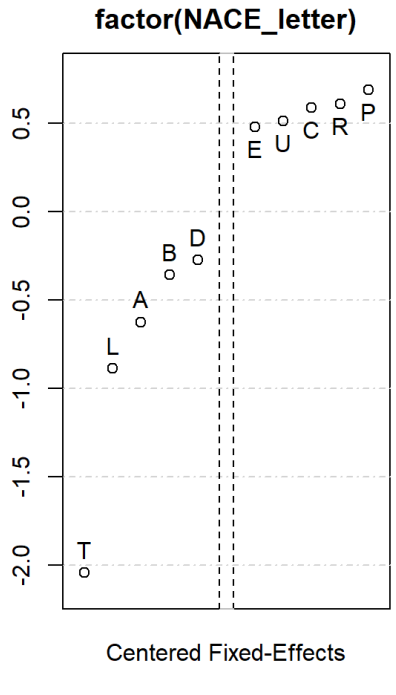
LPM for inclusion in sample

Fixed effects regression for TL3 pairs

South West most likely to be in the sample when controlling for firm size and sector.

Some areas of Scotland, London, and North-West least likely





Education; Arts and entertainment; Manufacturing

Activities of households; Real Estate; Agriculture